

Reaction dynamics analysis of a reconstituted *Escherichia coli* protein translation system by computational modeling

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Edited by Michael Levitt, Stanford University School of Medicine, Stanford, CA, and approved January 11, 2017 (received for review September 14, 2016)

To elucidate the dynamic features of a biologically relevant large-scale reaction network, we constructed a computational model of minimal protein synthesis consisting of 241 components and 968 reactions that synthesize the Met-Gly-Gly (MGG) peptide based on an *Escherichia coli*-based reconstituted in vitro protein synthesis system. We performed a simulation using parameters collected primarily from the literature and found that the rate of MGG peptide synthesis becomes nearly constant in minutes, thus achieving a steady state similar to experimental observations. In addition, concentration changes to 70% of the components, including intermediates, reached a plateau in a few minutes. However, the concentration change of each component exhibits several temporal plateaus, or a quasi-stationary state (QSS), before reaching the final plateau. To understand these complex dynamics, we focused on whether the components reached a QSS, mapped the arrangement of components in a QSS in the entire reaction network structure, and investigated time-dependent changes. We found that components in a QSS form clusters that grow over time but not in a linear fashion, and that this process involves the collapse and regrowth of clusters before the formation of a final large single cluster. These observations might commonly occur in other large-scale biological reaction networks. This developed analysis might be useful for understanding large-scale biological reactions by visualizing complex dynamics, thereby extracting the characteristics of the reaction network, including phase transitions.

computational modeling | protein translation | cell-free protein synthesis | quasi-stationary state | network analysis

Biological systems contain and are driven by multiple molecular components and reactions that form a complex reaction network. To elucidate the fundamental rules or features underlying complex reaction networks, various abstract computational models have been constructed (1–4). For example, cellular automaton (2) and NK models (1) have been applied to extract the fundamental features in the patterning and evolution of biological systems, respectively. A simplified cellular model was used to reveal why expressed gene copy number follows the power-law distribution (Zipf's law) (3). Although these models capture only the basic features of biological systems, they have helped elucidate the complex properties of life mechanisms. Additionally, although models that enumerate all detailed cellular processes have been constructed (5–7), attempts to extract the features or rules from such detailed models have been performed rarely.

A single enzymatic reaction can often be described by Michaelis-Menten kinetics, but once reactions are connected to one other, it becomes difficult to understand and capture a complete description of the reaction dynamics because of its high dimensionality. Despite such complexity, many high-dimensional dynamic data obtained from a variety of numerical models exhibit common behavior (reviewed in ref. 8). The models in the phase space show itinerancy over several quasi-stationary states (QSS) (9–12). For example, in a closed system that includes a random catalytic reaction network with

reversible reactions, all reactions are not always in a dynamic state in which the concentration of components changes over time. The concentrations of some components will be constant and appear to be in a QSS even though the entire reaction network has not yet reached final equilibrium. The components in a QSS subsequently change their concentration and reach another QSS (4). The appearance of successive and multiple QSS is crucial for peptide oligomerization (13) and the formation of epigenetic cellular memory through multisite modification of proteins (14). In contrast to these artificial model-based studies (4) or relatively small (~10 component) biochemical reactions (13, 14), limited information is available about the occurrence of QSS in detailed and biologically relevant large-scale models. The existence of multiple QSS implies that the reaction progresses through multiple different reaction phases; thus, consideration of the QSS might emphasize the dynamics of large-scale reactions as they move toward the final equilibrium or steady state. Elucidating QSS emergence and its dynamics in naturally existing large-scale reaction networks might provide novel insights into the dynamics of high-dimensional biological networks.

Here, we focused on the protein synthesis system, one of the most important and central large-scale catalytic reactions in cells, and we investigated the occurrence of components in a QSS (QSS components). The arrangement of components was then mapped in the whole reaction network structure, and time-dependent changes were observed, such as where and when the QSS components appear and disappear in the network structure.

Significance

Biological systems are driven by multiple components and interactions that form a complex reaction network. We developed a method to analyze their dynamics by focusing on the component in temporal plateaus, or a quasi-stationary state (QSS). The analyses, using a computational model of a minimal in vitro protein synthesis system, showed that components in a QSS form clusters that grow over time. However, the growth is not in a linear fashion: The process involved collapse and regrowth of the formed clusters, where the collapse was closely related to the phase transition in the reaction network. These observations suggest that the studies focusing on the QSS might be useful for understanding of complex reaction dynamics.

Author contributions: T.M. and T.Y. designed research; T.M. and Y.S. performed research; T.M., N.T., and Y.S. contributed new reagents/analytic tools; T.M., K.H., and Y.S. analyzed data; and T.M. and Y.S. wrote the paper.

The authors declare no conflict of interest.

This article is a PNAS Direct Submission.

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This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1615351114/-DCSupplemental.

We started by constructing a large-scale kinetic model of protein translation based on experimental data, an *Escherichia coli*-based reconstituted cell-free protein synthesis system (15). The constructed computational model, which can synthesize a Met-Gly-Gly peptide, consisted of 241 components and 968 reactions. By incorporating the kinetic parameters collected from the literature and component concentrations from the experimental data in the simulation, the presence of components at QSS in a logarithmic time scale was revealed. Furthermore, we found that small clusters formed by the QSS components emerged in the early phase of the reaction and gradually link with each other in later phases to ultimately form a single large cluster. Cluster growth was not linear but involved the collapse and regrowth of clusters. The observed dynamics of the QSS components might be a general feature of large-scale biological reaction networks.

Results

Construction of a Minimal Component Computational Model. In this study, we constructed a deterministic model of mRNA-directed protein synthesis based on the components of an *E. coli*-based reconstituted in vitro translation system (IVT), the PURE system (15). The PURE system integrates only highly purified components in a test tube to facilitate in vitro protein synthesis. Omitting any component reduces or halts protein synthesis activity, which experimentally demonstrates that the included components are the minimum required components for *E. coli*-based protein synthesis. The computational model was constructed according to the generally understood molecular mechanisms of the *E. coli* protein translation process and a number of previous kinetic studies (*SI Results, Model Construction*). Ordinary differential equations were used to describe the rates of all of the reactions, and the reactions were modeled as either first- or second-order reactions (*Datasets S1–S26*). Finally, the full, constructed model was implemented in Matlab (Mathworks) for simulation studies.

The full model was constructed to synthesize a formyl-Met-Gly-Gly tripeptide (fMGG) (*Datasets S1–S26*). Note that the model can be expanded to synthesize other peptides (*Discussion*). The constructed model included 241 components, 27 of which are the initial components. For analytical purposes, 241 components were classified into five functional groups: initiation, elongation, aminoacylation, termination, or energy regeneration (Fig. 1*A*). The model included 968 reactions, all of which were modeled as reversible reactions except for the degradation pathways, which were modeled as irreversible reactions.

The kinetic parameters simulating protein synthesis in the constructed model were obtained primarily from the literature (Fig. 1*B* and *Dataset S27*). According to how the parameters were assigned, they were classified into five groups: publication-based, assumed fast, assumed slow, hydrolysis, and degradation (Fig. 1*B*). Approximately 40% were assigned based on values reported in the literature (*SI Results, Parameter Assignment*). Although there were many assumptions and hypotheses made in this process, to our surprise, these parameters generated reasonable simulation results, as shown below.

fMGG Tripeptide Synthesis Simulation. The concentrations of the 27 initial components from the optimized PURE system, which we recently developed, were used as input (16) except for creatine kinase (CK) and nucleotide diphosphate kinase (NDK), and all of the other intermediates and products were set to zero (*Dataset S27*). The concentrations of CK and NDK were increased by 30-fold to maintain a high biochemical energy level (Fig. *S1*).

The simulation was performed up to 1,000 s, which takes about 2 s in a standard laptop computer. The time course of fMGG tripeptide synthesis exhibited a concave curve in the presteady state (<30 s), which is typical of a multistep reaction, and then increased constantly over time (Fig. 2*A*). A lag time of less than a minute is often observed experimentally for cell-free protein synthesis systems

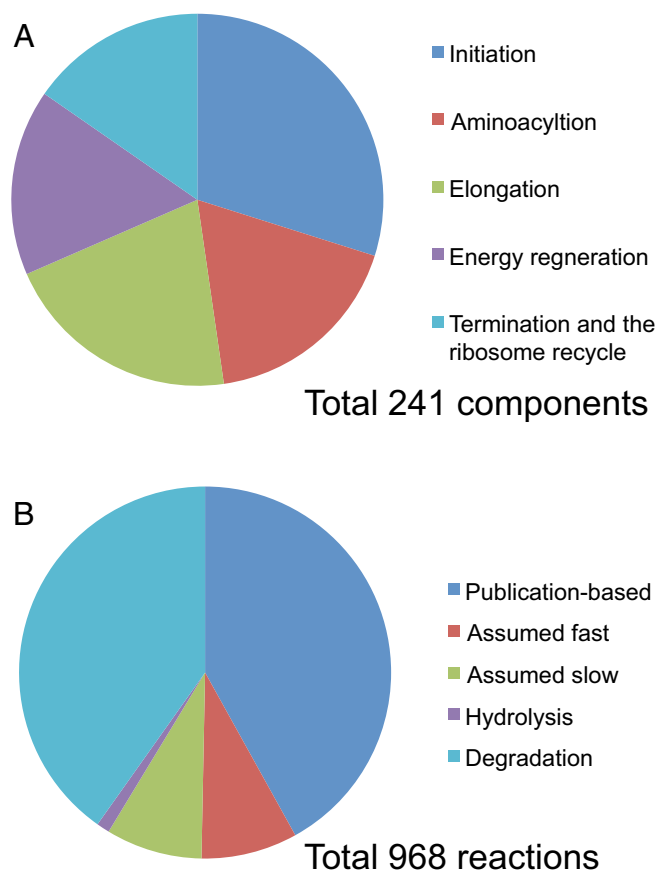


Fig. 1. Components and parameters of the minimal protein synthesis model. (A) Functional classification of 241 components constituting the model. A detailed list is found in *Dataset S27*. (B) Classification of 968 kinetic parameters according to the source. A detailed list is found in *Dataset S27*.

(17). Furthermore, the synthesis rate of the fMGG tripeptide was 0.15 (amino acids per second per mRNA), which was in the same range as the original PURE system (0.08 amino acids per second per mRNA) (16).

Each reaction module was simulated to validate the adequacy of the model and the parameters, and the behaviors of each module were similar to that observed experimentally (Fig. *S2* and *SI Results, Validation of the Constructed Model and the Assigned Parameters*). To further evaluate the computational model, the effect of varying the concentrations of the 27 initial components from 1/100- to 100-fold of the default value on fMGG tripeptide synthesis was investigated (Fig. *S3A* and *B*). The observations can be interpreted on the basis of the constructed model (*SI Results, Validation of the Constructed Model and the Assigned Parameters*). In addition, as ATP and GTP are used at multiple steps of the protein synthesis, the rate of fMGG tripeptide synthesis is expected to increase exponentially with increasing concentrations of ATP or GTP, which was the case in our model (Fig. *S3C*). These observations suggest that the present model reflects the characteristics of typical biochemical reactions.

Analysis of the QSS Components. The time development of the concentration of all of the components was plotted as a double-logarithmic plot (Fig. 2*B*). When the fMGG tripeptide synthesis rate became nearly constant after ~100 s (Fig. 2*A*), the concentrations of ~70% of the components (see below for details) reached a plateau (Fig. 2*B*). This result suggests that fMGG synthesis reaches a final steady state and that the peptide synthesis rate becomes constant within a few minutes. We also found that

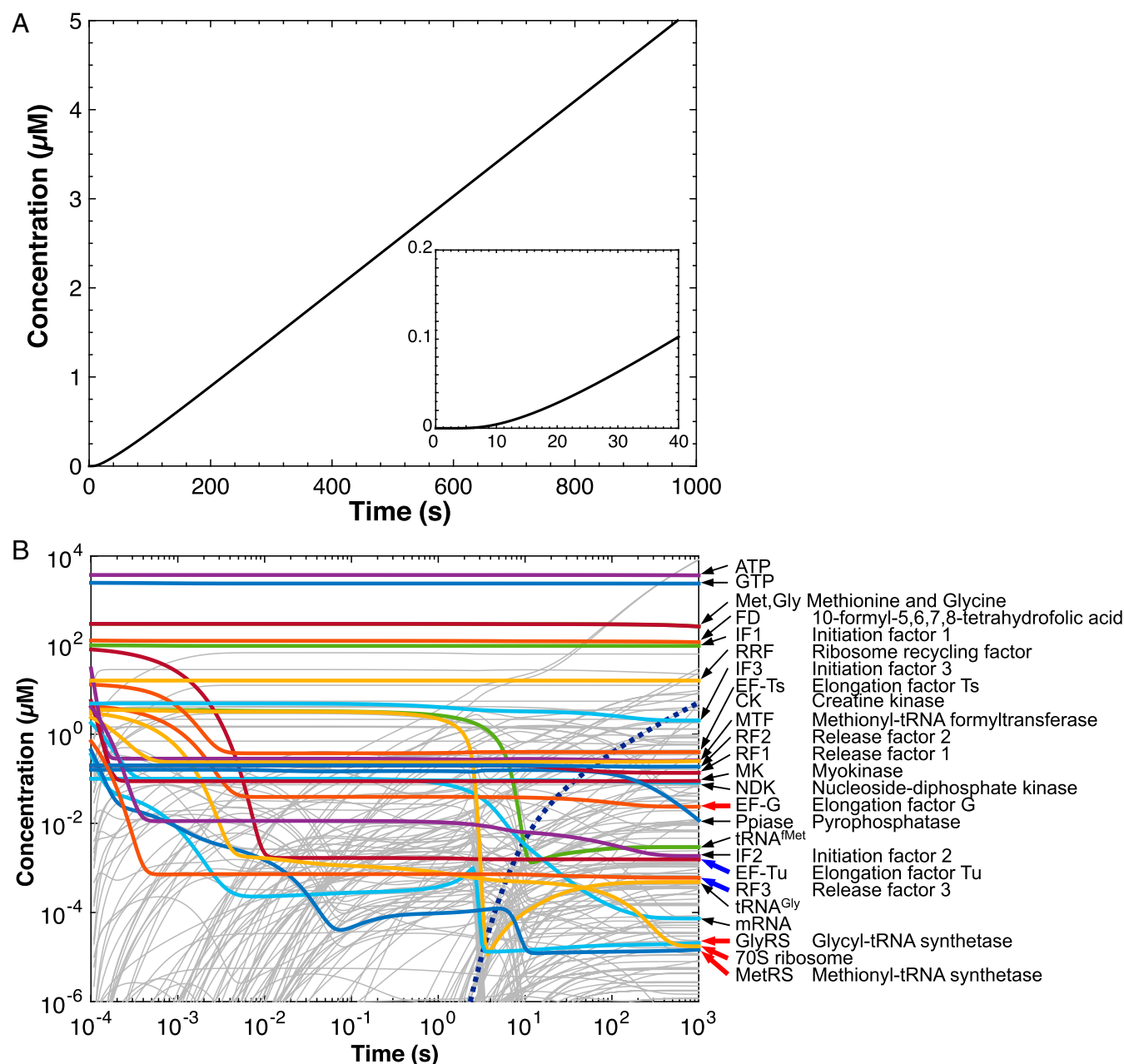


Fig. 2. fMGG tripeptide synthesis simulation. (A) Time course of the fMGG tripeptide over a linear scale. *Inset* shows the time-course data up to 120 s. (B) Time course of the 241 components in a double logarithmic plot. The data from 27 initial components are presented as bold lines with color, and the data for the fMGG tripeptide are presented as a blue dashed line. The concentration change in creatine phosphate (CP) is not shown, given that its initial concentration was 50 mM and did not decrease enough to appear on the vertical axis scale. The abbreviations of the initial components are listed on the right. The simulation was performed using the kinetic parameters and initial concentrations provided in [Dataset S27](#). Some of the components exhibiting fluctuation over several plateaus are indicated by red arrows, and some components that reach a plateau and remained constant until 1,000 s are indicated by blue arrows. Detailed data are found in [Dataset S28](#).

some components achieved a plateau before reaching the final plateau (Fig. 2B, red arrows). For example, the time course for elongation factor-G (EF-G) reached a plateau at 4 ms that continued up to 20 s. Then, the concentration changed and reached the final plateau, which started at ~ 400 s. Similarly, the 70S ribosome, Methionyl-tRNA synthetase (MetRS) and Glycyl-tRNA synthetase (GlyRS) concentrations also reached a plateau before the final plateau, but the exact timing of the first and last plateaus were different among the components. In contrast, some components reached a plateau and remained constant until 1,000 s; however, the timing was different between these components as

well [e.g., elongation factor-Tu (EF-Tu) and release factor (RF)3, marked with blue arrows in Fig. 2B].

When the above plateaus are interpreted as QSS, the components in the reaction network might transition between the QSS and the non-QSS at different times, with $\sim 70\%$ of the components eventually reaching the QSS at 1,000 s. This finding resembles the phenomena proposed by Kaneko and Tsuda (8). However, when and where the QSS components appear and disappear in the network structure and how the network of QSS components grows over time remains unclear. Therefore, we decided to focus on the QSS and mapped QSS components on the reaction network

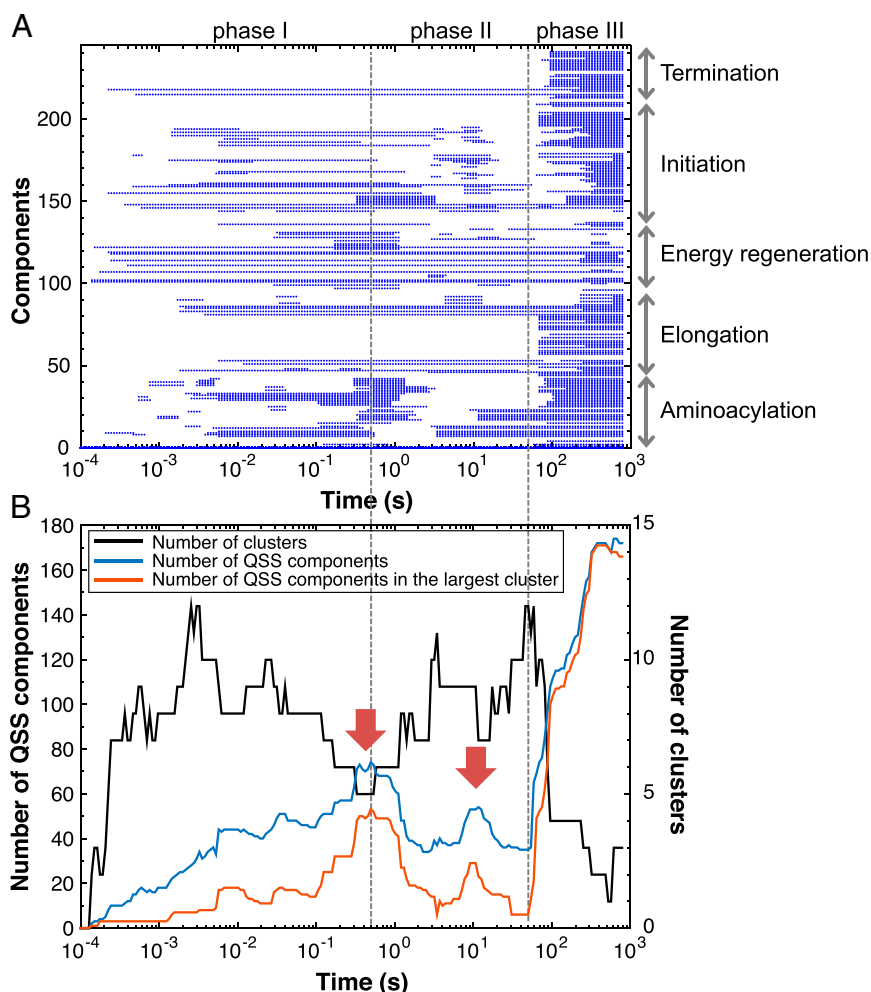


Fig. 3. QSS analysis. (A) The time at which the component is at QSS is shown. A blue dot was plotted when each of the 241 components (vertical axis) is at QSS (horizontal axis). On the right, the functional groups in which each of the components was classified are presented (Dataset S27). Detailed data are found in Dataset S28. (B) QSS occupancy. The number of QSS components (blue line) at each time point is plotted. Two arrows indicate the two major peaks described in the text. The number of clusters (black) at each time point is provided in the right axis. The time was classified into three phases, phase I to III, depending on the characteristic dynamics (see text for details).

structure using a developed representation scheme, which might provide novel insights into the dynamics of this reaction network. We defined the QSS components at each time point, analyzed the growth of the network structure containing the QSS components, and discovered features describing the network growth dynamics of a large-scale reaction.

First, from the time-course data, the slope of concentration change for each component, which was calculated as the slope of four consecutive time points in the double-logarithmic plot, was obtained for 200 time points logarithmically spaced between 10^{-4} and 10^3 s (Fig. S4). Second, other than some exceptions described below, when the absolute value of the calculated slope was less than 0.2 [$\Delta \log(c)/\Delta \log(t)$; where c represents concentration (μM) and t represents time (s)] for more than three consecutive time points, the component was defined as being in a QSS (Fig. 3A and Fig. S4). If the slope was below 0.2 because of the initial components maintaining the initial state with little concentration change, the component was not considered as being in a QSS (e.g., Met and Gly in Fig. 2B). As shown in Fig. 3B, the number of QSS components increased over time. At 500 s, $\sim 70\%$ of the components (173 of 241) reached a QSS. Among the 68 non-QSS components at 500 s, 29 are degradation products and were not considered in this study by setting the rate constant of degradation to zero (SI Results, Parameter Assignment).

Note that small changes in the threshold value did not affect the dynamics of the number of QSS components (Fig. S5).

Despite the trend that the number of QSS components increased over time (Fig. 3B, blue line), the increase was not monotonic. Two major peaks were observed in the time course data: one at 0.5 s and another at 10 s (Fig. 3B, red arrows). According to the major peaks, we classified the time course in three phases, from phase I to III (Fig. 3). In phase I, the components involved in aminoacylation, initiation, and energy regeneration began to reach QSS within a few milliseconds, and the total number of QSS components increased until 0.5 s. In phase II, the components involved in aminoacylation and energy regeneration moved away from the QSS, resulting in a decreased number of QSS components and generating the first major peak at ~ 0.5 s (Fig. 3B). Then, another small major peak appeared at 10 s, after which the total number remained relatively unchanged up to 500 s. In phase III, most of the components began to reach QSS. Consequently, the system reached the final steady state at 500 s.

Topological Arrangement and Temporal Dynamics of QSS Components in the Reaction Network. We next mapped the QSS components in the reaction network structure and investigated its temporal dynamics, such as when and where the QSS components appear in the reaction network structure, whether the components assemble to form clusters

and, if so, how the clusters grow over time. We first defined nodes and edges in the entire reaction network (Fig. 4A). The components that were at QSS were considered nodes. If two components appeared simultaneously in the rate reaction more than once in the 968 reactions, the two were considered connected; thus, an edge was drawn between the two (Fig. 4A). In this way, the data at each time point were converted into a graph (Fig. S6). An example of the graph at 0.87 s is shown in Fig. 4B, depicting six clusters. The number of clusters (Fig. 3B, black line) and the number of nodes in the largest cluster (Fig. 3B, orange line) were calculated for each time point. The number of clusters repeatedly increased and decreased over time. Finally, at 500 s a single cluster appeared.

To visualize the development of the network structure, pie charts were used (Fig. 4C). In the early stage of phase I, small clusters containing nodes with a single functional group appeared (0.0013 s). The cluster size then grew by absorbing nodes belonging to other functional groups (0.17 and 0.39 s). During phase II, the size of the cluster became smaller, and simultaneously the number of clusters consisting of nodes with one or two functional groups increased (compare 0.39 and 4.4 s in Fig. 4C). This phase corresponded to the collapse of the cluster. Subsequently, the size of the cluster grew (4.4–9.9 s) and collapsed again (9.9–49 s). Finally, in phase III the cluster size grew, and a large cluster consisting of components other than energy regeneration appeared (112 s). Then, the cluster continued to grow by absorbing energy regeneration nodes, resulting in a single cluster with 173 components (567 s).

Using a new representation scheme, we succeeded in visualizing the development of the QSS during protein synthesis and found that the cluster of QSS components grows over time but not linearly. The development involves a collapse of the network structure (i.e., simultaneous decrease in the cluster size and increase in the number of clusters). The observed two collapses correspond to the two major peaks in the number of QSS components at 0.5 s and 10 s (Fig. 3B). In the next section, we describe the mechanism of the observed collapse and regrowth.

The Mechanism of Cluster Collapse. The peak at 0.5 s can be explained by the conversion of free tRNAs to aminoacyl-tRNAs. At 0.5 s, concentrations of free tRNA^{Met} and tRNA^{Gly} begin to decrease because of their conversion to aminoacyl-tRNAs by the aminoacylation network (Fig. 5A, bold lines). Following the first rapid decrease in free tRNA concentration, many of the aminoacylation-related components moved away from the QSS and then again reached the QSS during phase II and phase III (Figs. 3A and 5A). From these observations, we hypothesized that free-tRNA depletion triggered the reduction in the number of QSS components. Indeed, when the simulation was performed with a 10-fold increased concentration of tRNA^{Met} and tRNA^{Gly}, the timing of QSS component decrease was delayed (Fig. 5B), supporting the hypothesis that the peak at 0.5 s arises from the depletion of free tRNAs. These results also imply that the way free tRNA is supplied to the aminoacylation network has changed before and after the

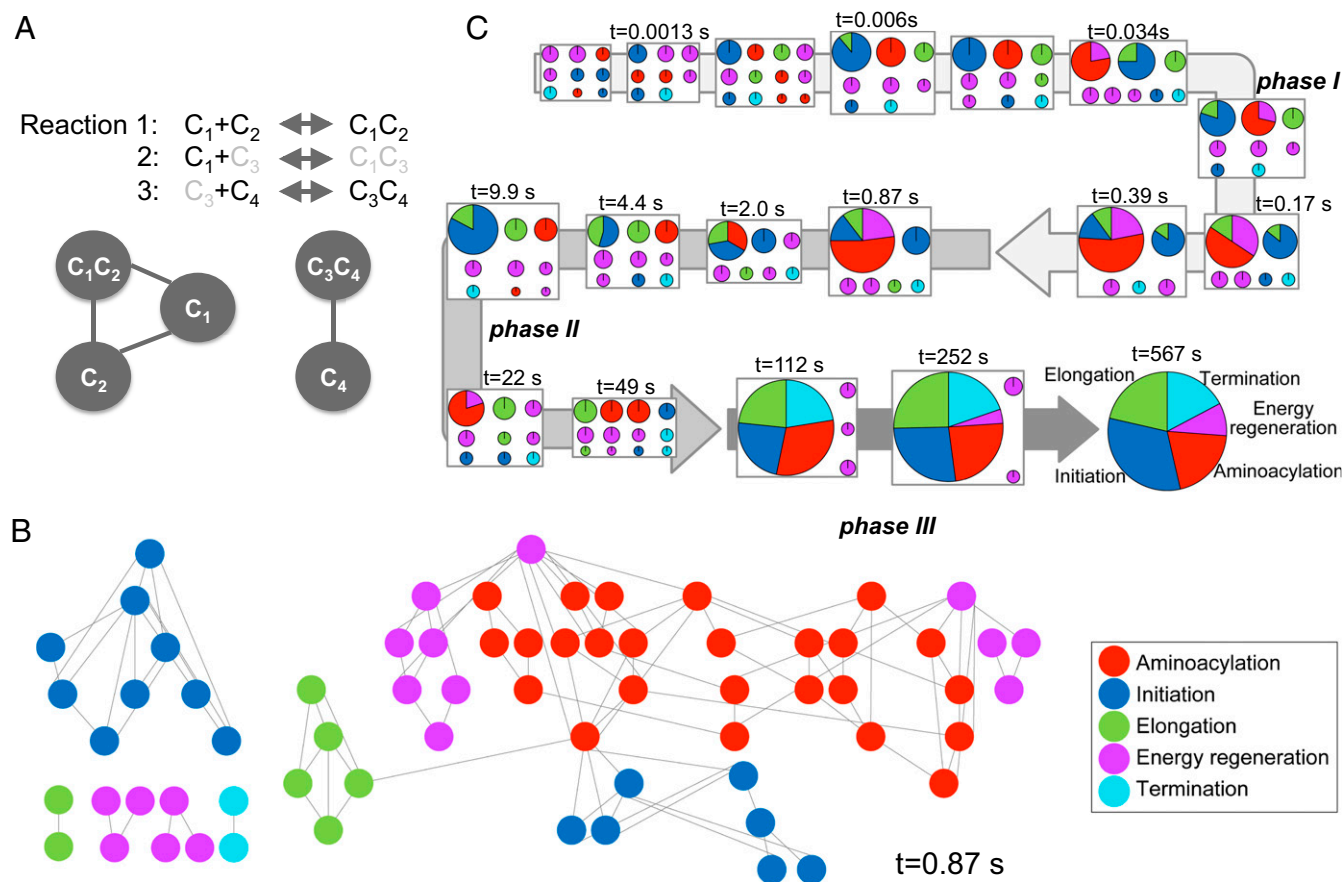


Fig. 4. QSS network analysis. (A) The schematic of the reaction network analysis. Let us consider a system consisting of three reactions and seven components, in which at time t , C_3 and $C_1 C_3$ (gray letters) are not at QSS, whereas other components are at QSS (black letters). Components at QSS are considered nodes, and an edge is drawn when two nodes are present in greater than one reaction scheme. At time t , two clusters are present with the largest cluster having three nodes. (B) Network of the QSS components at time 0.87 s is shown. Each component is colored according to its functional classification. More examples are presented in Fig. S6. Detailed data are found in Dataset S29. (C) The dynamics of the network structure. Each pie chart shows clusters in which the size represents the number of nodes that constitute the cluster, and the color represents the functional groups.

peak. Before the peak, the network uses free tRNAs supplied at the beginning of the reaction. After the peak, the network uses free tRNAs supplied from translation system: that is, tRNAs once used in the translation system are released after use, and then recycled for aminoacylation.

The peak at 10 s can be described by the presence of a slow reaction among the initiation reactions (Fig. 5C). The dissociation of the 70S ribosome (designated as RS70S) to the 50S and 30S ribosomal subunits is a very slow reaction relative to the others. As seen in Fig. 5C, although the RS70S concentration decreases at $\sim 10^{-2}$ s, a substantial fraction of it is converted to RS70S_IF1 (Fig. 5C, blue dashed line), which exhibits a constant concentration up to ~ 10 s. In addition, RS70S_IF1_IF3 (Fig. 5C, orange dashed line), one of the most abundant intermediates, accumulated slowly over time and reached its QSS at ~ 10 s. These results indicate that the dissociation of the 70S ribosome to the 50S and 30S subunits is a very slow process. Because of this bottleneck, many of the downstream components (RS70S_IF1_IF3) reach a QSS at ~ 10 s followed by a decrease in concentration at ~ 100 s as a result of the flux toward the downstream components. Indeed, when the reaction was initiated using a dissociated 70S ribosome (i.e., 50S and 30S ribosomal

subunits), the small peak disappeared and the increase in the number of QSS components occurred earlier (Fig. 5B).

To identify and confirm the bottleneck reactions during the presteady state and clarify its relationship to the peak at 10 s, we examined the effect of decreasing each kinetic parameter with the value more than zero on the fMGG synthesis. We found that by decreasing the parameter by 10-fold, only 28 among 483 affected the yield of fMGG peptide by more than 10% at 1,000 s and only 20 affected the lag time of fMGG synthesis, defined as a time required to reach fMGG concentration of $0.02 \mu\text{M}$, by more than 10%, indicating the robustness of the network (Dataset S28). Lag times represent the time required for the reaction to reach the final steady state: that is, the duration of presteady state. Parameter changes affecting the lag time indicate that the corresponding reactions are the bottleneck reactions. Note that 10-fold decrease of any parameter, which was assumed to be fast, did not affect the yield nor the lag time, confirming that the assumed values are fast enough to prevent the corresponding reactions from being rate-limiting. Furthermore, the analysis of the lag time showed that 10-fold reduction of 447 parameters do not affect the lag time,

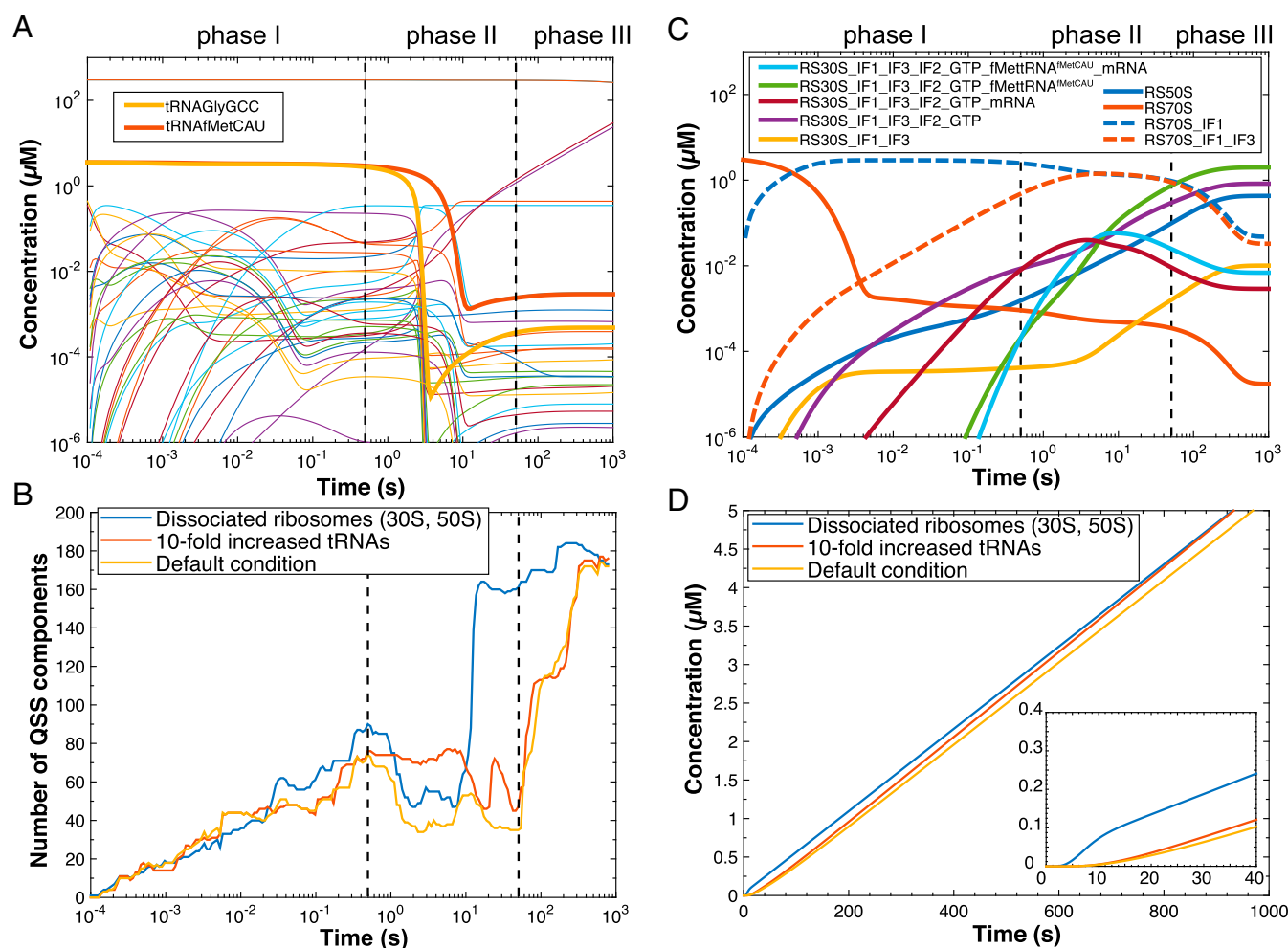


Fig. 5. Mechanism of cluster collapse. (A) Time course of the concentrations of the aminoacylation-related components among the 241 components in a double logarithmic plot. The concentrations of free tRNAs that start to decrease at ~ 0.5 s are highlighted. Detailed data are found in Dataset S28. (B) Time course of the number of QSS components for different initial conditions. Default condition: the values given in Dataset S27 were used. Dissociated ribosome: the concentration of the 70S ribosome was set to zero, and the concentrations of both the 30S and 50S subunits were set to $3 \mu\text{M}$, whereas the others remained unchanged. Ten-fold increased tRNAs: the concentrations of tRNAs were increased by 10-fold simultaneously, whereas the other components remained unchanged. (C) Time course of the concentrations of the initiation reaction-related components on a double logarithmic plot. The abbreviations of the components shown are described in Dataset S27. (D) Time course of fMGG tripeptide synthesis on a linear scale. Inset shows the time course up to 120 s.

suggesting that the corresponding 447 reactions are at least 10-fold faster than the slowest reaction (Fig. S7A).

Two parameters were found to increase the lag time by more than twofold. These two are likely to be the bottleneck reactions during the presteady state. One is responsible for the 70S initiation complex formation by 50S subunit binding to the 30S initiation complex and the other is the subsequent GTP hydrolysis on the 70S initiation complex (Dataset S4; reactions 529 and 717 in Dataset S27, respectively). The former is the second-order reaction that is affected by the free 50S subunit concentration and the latter is the first-order reaction that proceeds on the 70S ribosome. The peak at 10 s in the QSS occupancy was shown to be slightly higher, broadened, and delayed (Fig. S7B), when these two parameters were reduced, suggesting these bottleneck reactions strongly correlate with the emergence of the peak.

We also found two parameters that decrease the lag time by more than 1.5-fold. One is responsible for the side reaction in which EF-G/GTP complex binds to the free 50S subunit (Dataset S7; reaction 298 in Dataset S27) and the other is the reverse reaction of the 70S initiation complex formation by 50S subunit binding to the 30S initiation complex (Dataset S4; reaction 530 in Dataset S27). The decrease in the earlier parameter is likely to increase the free 50S subunit concentration. The latter parameter is the reverse reaction of the lag-time increased parameter (reaction 529 vs. reaction 530 in Dataset S27). When these parameters are decreased, the peak at 10 s in the QSS occupancy disappears, similar to what we observed by initiating the reaction using a dissociated 70S ribosome (Fig. S7C). These studies suggest that the bottleneck reactions are the cause of the peak appearance.

We note that the emergence of peaks does not necessarily depend on the final product synthesis. When the GTP hydrolysis of EF-G on the translating ribosome (Dataset S8) is deactivated (i.e., the kinetic parameter for reaction 22 in Dataset S27 is set to zero), the above two peaks remain and fMGG tripeptide is not synthesized (Fig. S7D). When the ATP hydrolysis of aminoacyl-tRNA synthetases (ARS) (Datasets S17–S20) is deactivated (i.e., the kinetic parameters for reactions 140, 165, 197, and 239 in Dataset S27 are set to zero), only the first peak at 0.5 s disappears and fMGG tripeptide is not synthesized (Fig. S7D).

Diverse QSS Occupancy Results in a System with Similar Activity. In the previous section, we found that altering the initial component concentrations affects the time course of the number of QSS components and QSS occupancy (Fig. 5B). When we investigated fMGG tripeptide synthesis with identical changes to the initial condition, we observed little effect (Fig. 5D), suggesting that diverse QSS occupancy might still generate a system with an approximately identical peptide synthesis rate.

To test this idea more systematically, we used the results of a simulation performed by altering the concentrations of the initial components individually (Fig. S3). Among all of the time-course data, we chose cases in which the amount of fMGG tripeptide synthesized at 1,000 s was less than 10% different from the default setting (116 conditions from examined 216 conditions). The QSS occupancy of the chosen datasets was then compared with the results of the default settings. As a result, the presence of diverse transitions, where the coefficient of variation reached 50.0% at most was observed (Fig. 6 and Fig. S8), indicating that a system with similar performance can be achieved despite a variety of QSS occupancies.

Discussion

Here, we constructed a computational model of protein synthesis based on the components of the PURE system, and simulated the reaction using parameters collected primarily from the literature. We found that the reaction achieves a steady state in less than 1 min, and the component concentration transitions between QSS and non-QSS before achieving a final steady state. By focusing on the QSS, we found that the small clusters formed by

the QSS components that emerged in the early phase of the reaction gradually link with each other in the later phase to finally form a single large cluster. Cluster growth was not linear; it involved the collapse and regrowth of clusters, as shown by the integrated figure using QSS occupancy and pie charts (Fig. S9).

Based on the excellent control of the input conditions, IVT has been used as a template for computational modeling (18–20). Coarse-grained models consisting of a limited number of parameters and reactions that consider protein translation as a single reaction step have been developed and used to understand the dynamics of protein translation. As a result, kinetic parameters and adequate models have been successfully developed that can predict the dynamics of protein synthesis in IVT with high accuracy (18, 19). However, unlike the model presented in this study, the coarse-grained model does not provide information about the dynamics of individual components.

Detailed dynamics of individual component concentrations were obtained from our minimal protein translation model. We then focused on components at QSS, when the component concentration becomes almost constant. When counting the number of QSS components at each time point, we identified two major peaks (Fig. 3B). The presence of peaks indicates the collapse of the QSS cluster and is likely among the general properties of large-scale reactions. The first peak was caused by free tRNAs conversion to aminoacyl-tRNAs (Fig. 5A). At this time point, the way free tRNA is supplied to the aminoacylation network has changed: before the peak free tRNA is supplied from the initial material, whereas after the peak it is supplied from the translation system (i.e., recycled material). Other examples of recyclable substrates, such as nicotine adenine dinucleotide, flavine adenine dinucleotide, and CoA might show similar peaks in the QSS occupancy. Thus, the collapse and regrowth of QSS clusters before achieving a final steady state is likely to occur with other large-scale reactions.

The second peak was caused by the slow dissociation of the 70S ribosome to the 50S and the 30S subunits (Fig. 5C), and the presence of such a bottleneck in the reaction network caused another major peak. The accumulation of a particular component because of a slow downstream reaction and its decrease thereafter will result in a peak. The above two phenomena occurred for protein synthesis, the change in the supply of recyclable substrates and reaction cascade bottleneck, might also occur for other large-scale biological reaction networks, such as glycolysis or subsequent citric acid cycle; thus, the presence of peaks is likely a general property of large-scale reactions.

The significance of the peak and the collapse of the network structure are discussed below. In our protein translation model, there is a phase transition in the reaction network before and after the two major peaks. Before the major peaks in phase I, tRNAs used in the reactions are supplied from free tRNAs, and the ribosomes are in the 70S conformation. The number of QSS components increases almost linearly during this phase. During phase II, tRNA depletion occurs, and the bottleneck of the initiation reaction is dissolved, which had perturbed the growth of the network. After the two major peaks, in phase III, tRNAs released at the elongation or termination steps are used as inputs for the entire reaction, and the initiation step is no longer limited by the bottleneck. Thus, the peak and the collapse of the network structure correspond to the phase transition of the reaction. To date, a number of large-scale models, including a whole-cell computational model, have been constructed by incorporating the accumulated knowledge on genes, proteins, the interactions among them, and the catalytic reactions involved (5–7, 21, 22). These models have been used to determine whether the current knowledge is sufficient to describe the experimental observations and to predict the output of the biological system with untested inputs. These models are expected to facilitate new biological discovery. In addition to these previous analyses, the application of our methods to such large-scale models might help us understand the models by visualizing the dynamics, by clarifying the

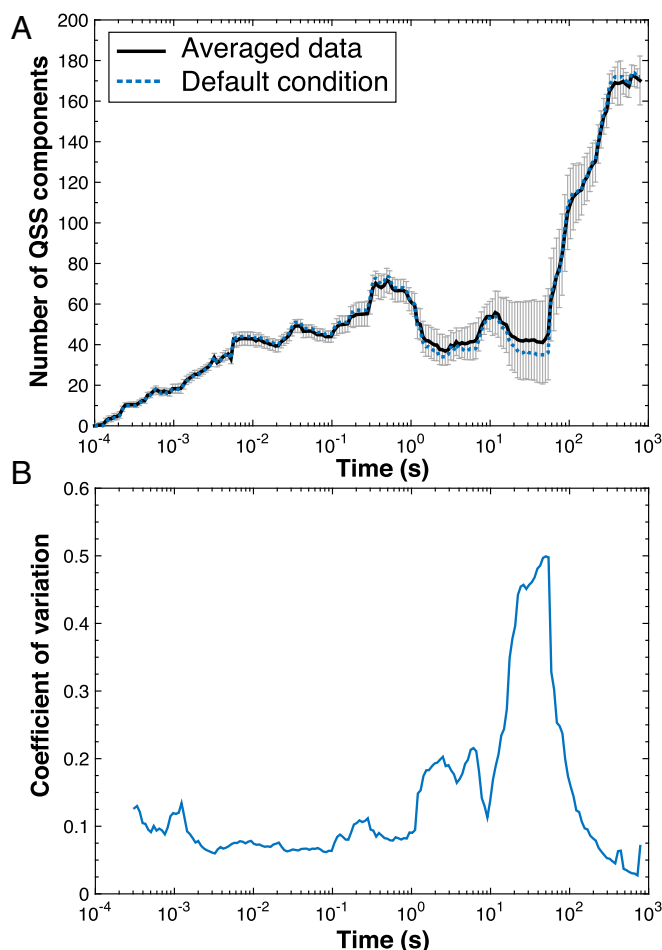


Fig. 6. The effect of altering the concentrations of the initial components on the QSS occupancy. Among the 216 initial datasets tested (Fig. S3A), 116 datasets, in which the amount of fMGG tripeptide synthesized at 1,000 s was less than 10% different from the original value, were chosen for analysis. (A) Average and SD at each time point is shown. (B) The time course of coefficient of variation (CV). The CV value at each time point of the 116 datasets shown in A was calculated. The result shows that CV shows a peak and reaches a maximum of 0.5 before the final steady state.

relationship among the components, and by extracting the characteristics of the reaction network including the phase transitions.

We also found that protein synthesis exhibits diverse QSS occupancy as it reaches the final steady state to generate a system with approximately identical fMGG tripeptide synthesis activity (Fig. 6). The presence of multiple trajectories is analogous to the protein-folding pathway in which the native structure is reached from the unfolded structure by sampling multiple conformations in the energy landscape funnel. Here, we have shown the presence of multiple trajectories during protein translation, which is likely to be a general property of large-scale kinetic reactions.

We showed the results of synthesizing a particular peptide, the fMGG tripeptide; however, this model can be expanded to study

the synthesis of peptides of other sequences and lengths. The current model consists of two amino acids (Met and Gly), two tRNAs (tRNA^{Met} and tRNA^{Gly}), and two ARSs (MetRS and GlyRS). The model can be expanded by including all amino acids and tRNAs. A correspondence table that describes which tRNAs read which codons can be implemented in the model to evaluate the effects of codon use that are mainly affected by each tRNA concentration.

The present study focused on presteady-state kinetics. It will be interesting to study steady-state and poststeady-state kinetics where degradation of components or physiological changes, such as pH decrease, occur by the accumulation of by-products (23). Such studies might require the cut down of calculation costs for long-run simulations. Introduction of the steady-state approximation (i.e., Michaelis–Menten kinetics) or other approximations for the development of more coarse-grained model starting from the present model might be effective for more complicated and long-term reactions. In addition, the PURE system is a reconstituted system and therefore supports changes to any of the initial component concentrations to examine their effect on protein synthesis dynamics. Therefore, a detailed parallel evaluation of the experimental data and the computational model should be possible. Investigating the discrepancy and the consistency of the two should contribute not only to confirming the accuracy of the computational model and the kinetic parameters, but also for predicting unknown kinetic parameters. For example, NTPs and amino acid concentrations can be quantified experimentally (24), and these experimental data might be used to predict kinetic parameters. These studies might deepen our knowledge on the entire cell-free protein synthesis system for generating improved systems (25). More importantly, the constructed simulation could be used to study the fundamental properties of large-scale kinetic models and to elucidate the mechanism of how complex cellular reactions are maintained at a steady state.

Materials and Methods

Model Construction. First, we constructed a kinetic model of the subsystems that constitute the minimal mRNA-directed protein translation system. The system was divided into 26 subsystems (Datasets S1–S26). Each subsystem was described in SBML (Systems Biology Markup Language) format (26) and the subsystems were combined into a single SBML file (Dataset S27). The resulting full model was implemented in Matlab (Mathworks) using ordinary differential equations for the simulation studies. The rationales for constructing each subsystem are described in SI Results, Model Construction.

Simulation. All of the simulations and calculations were performed using Matlab (Mathworks). The initial component concentrations and kinetic parameters that were used in the study are provided in Dataset S27. The motivations for obtaining the kinetic parameters are described in SI Results, Parameter Assignment. Each functional module was evaluated individually to verify the model and parameters (SI Results, Validation of the Constructed Model and the Assigned Parameters). All of the files used for simulations can be obtained from our website (<https://sites.google.com/view/puresimulator>).

ACKNOWLEDGMENTS. This work was supported by a Grant-in-Aid Grants 25282239, 16H00767, and 15K12756 (to T.M.), 26710014 and 26640134 (to Y.S.), and 25650147 (to K.H.) from the Japan Society for the Promotion of Science; the strategic programs for research and development (President's discretionary fund) of RIKEN (Y.S.); an intramural Grant-in-Aid from the RIKEN Quantitative Biology Center (Y.S.); and ERATO project of Japan Science and Technology (T.M.).

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